

AI-POWERED SOLAR PV INSPECTION AUDIT REPORT

Facility: Helios One 50MW Solar Plant | Generated: September 29, 2026 - 06:00 UTC | Standard: IEC 62446-3
Thermographic Audit

Total Panels 28	Total Audits 30	Fault Rate 80.0%	Critical Incidents 22	Mean ΔT Gradient +30.9°C
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1. Photovoltaic Fault Distribution Breakdown

Fault Category	Detected Count	Relative Prevalence	Dominant Severity
Physical Damage	8	26.7%	Critical
Normal	6	20.0%	Normal
Hot Spot	2	6.7%	Critical
Micro Crack	2	6.7%	High
Dust & Soiling	2	6.7%	Medium
Bird Dropping	2	6.7%	Medium
Discoloration	2	6.7%	Medium
Cell Defect	2	6.7%	High
Partial Shading	2	6.7%	Medium
Electrical Anomaly	2	6.7%	Medium

2. Critical & High Priority Action Register

Panel Code	Cell Grid	Fault Type	ΔT (°C)	Recommended Action Protocol
PNL-A1-02	R4 C1	Hot Spot	+58.2°C	Immediate thermal audit. Disconnect string from combiner box...
PNL-A1-03	R2 C6	Micro Crack	+23.4°C	Perform Electroluminescence (EL) imaging to check crack bran...
PNL-A1-04	R3 C7	Dust & Soiling	+17.4°C	Deploy automated or semi-automated cleaning robots with demi...
PNL-A1-05	R2 C2	Bird Dropping	+35.7°C	Targeted spot cleaning using biodegradable non-abrasive dete...
PNL-A1-06	R1 C10	Physical Damage	+50.2°C	De-energize module string immediately! Lockout/tagout (LOTO)...
PNL-A2-01	R6 C1	Discoloration	+23.1°C	Measure IV-curve fill factor to evaluate resistance changes....
PNL-A2-02	R3 C7	Cell Defect	+32.4°C	Perform anti-PID voltage reversal during nighttime if PID su...
PNL-A2-03	R2 C10	Partial Shading	+24.6°C	Trim surrounding vegetation. Re-calibrate tracker azimuth an...

Certification: This inspection report was automatically synthesized by the AI-Powered Solar Monitoring Deep Learning Pipeline in accordance with IEC 62446 Category 1 & 2 thermographic compliance specifications.